

Image2CAD

Image-to-CAD Jewellery Reconstruction

Development roadmap, research plan, results and validation strategy

Document status: Working technical plan

Current validated checkpoint: Ring01 Phase 3.3.2

Current scope: Rings, pendants, necklaces, ornaments, multi-stone jewellery and sculptural jewellery

Primary output: Editable, component-aware CAD with traceable evidence and validation reports

Focused follow-up: [High-Fidelity Five-View Jewellery Reconstruction](#) — depth, edge, texture, feasibility, compute plan and next validation milestones.

PDF edition: [Image2CAD project roadmap](#)

2D JEWELLERY TO EDITABLE 3D/CAD



1. Project objective

The project aims to reconstruct jewellery from one or more reference images while preserving both visible design details and editable jewellery structure.

The desired output is not only a visually similar mesh. It should identify and represent the jewellery as addressable parts:

- metal body, shank, shoulders, galleries, links and chains;
- every gemstone as an individual instance;
- stone cut, dimensions, pose and placement;

- every prong, bezel, channel, seat and cavity;
- engraved, pierced, embossed and sculptural details;
- named parameters that can change without rebuilding the complete model;
- STEP as the authoritative engineering exchange format, with STL, 3MF, OBJ or GLB as derived outputs.

For example, a ten-stone design should produce ten stable stone IDs. Removing or resizing one stone must update its seat, clearances and supporting metal without changing unrelated stones.

The long-term goal is a highly faithful and editable reconstruction. Ordinary uncalibrated images cannot prove exact hidden geometry or physical dimensions. The pipeline will therefore distinguish three levels of output:

1. **Visual reconstruction:** reproduces the appearance and silhouettes of the supplied views.
2. **Structural reconstruction:** contains plausible, individually editable jewellery components and valid topology.
3. **Manufacturing reconstruction:** adds calibrated scale, verified hidden geometry, minimum-wall and clearance rules, and physical inspection or scan evidence.

No score from the first level will be presented as proof of the third.

2. Current phase architecture

Phase	Purpose	Required output	Current state
Input gate	Validate image count, naming, focus, coverage and provenance	Input manifest, hashes, view map and quality warnings	Implemented in the upload auditor; capture calibration remains optional
Phase 1	Preserve and normalize image evidence	Foreground, crop transform, contours, edges, holes, symmetry and pixel measurements	Validated on Ring01 and all 120 dataset views
Phase 2	Extract foreground and visible component proposals	Jewellery, stone, metal, setting, shank, support/prong and shadow masks	Machine extraction validated; not human ground truth
Phase 2.2	Enforce local topology and cross-view consistency	Refined masks, amodal stone proposals and consistency report	Validated as a reconstruction initializer
Phase 2.3	Preserve arbitrary fine detail and require semantic review	Evidence graph, instance IDs, cross-view track hypotheses and review decisions	Implemented; real dataset review is pending
Phase 3	Build coarse multi-view geometry	Non-metric visual hull or coarse semantic reconstruction	Ring01 and dataset-wide experimental outputs exist
Phase 3.3	Fit valid exact solids against every view	Editable STEP, component solids, rendered comparisons and strict report	Ring01 numerical and topology targets pass
Phase 3.3.1	Make prongs individually addressable and smoothly curved	Four stable cubic claw IDs in one valid metal B-rep	Valid; image fit preserved
Phase 3.3.2	Correct inspection-view ambiguity	Reference-fitted previews in which all four prongs are visible	Valid; geometry unchanged

Phase	Purpose	Required output	Current state
Phase 3.4	Fuse camera, depth and normal evidence	Calibrated cameras, aligned depth, normals, confidence and uncertainty volume	Planned benchmark; human truth and controlled capture required
Phase 3.5	Reconstruct material, texture and fine surface detail	PBR maps, relief/displacement and source-view confidence atlas	Planned research phase
Phase 4	Build a universal component and constraint engine	Editable stones, seats, cavities, supports and dependency graph	Next major implementation phase
Phase 5	Reconstruct free-form and sculptural detail	Relief/displacement evidence converted into constrained surface detail	Planned research phase
Phase 6	Validate manufacturing constraints	Metric, topology, clearance, wall-thickness and casting report	Blocked until scale and manufacturing rules are supplied

Earlier checkpoints stay immutable. A new experiment receives a new version, and a rejected experiment is recorded before backtracking.

3. What has been implemented and measured

3.1 Phase 1: deterministic image evidence

OpenCV is used for border-background estimation, normalization, Canny edges, contours, holes, connected components, morphology, symmetry and pixel-space measurements. Phase 2.3 also records internal edges, ridge and valley evidence using top-hat and black-hat transforms, local relief with Laplacian responses, negative spaces and possible highlight interference.

Why OpenCV is used:

- it is deterministic, fast and light enough to run on every input;
- geometric measurements remain inspectable instead of being hidden in a model embedding;
- it provides fallbacks when a learned segmenter is unavailable;
- it is useful for validating learned predictions with independent contour and topology checks.

Result:

- all 120 views from 24 dataset objects were processed;
- each source is preserved and associated with generated masks, edges and measurements;
- the measurements remain in pixels because no metric calibration is available.

3.2 Phase 2: segmentation and normalization

The current stack combines SAM 2.1 Tiny, OpenCV and GrabCut.

- **SAM 2.1 Tiny** supplies promptable whole-object and local support proposals. It was chosen because it fits the laptop GPU and can generalize without a jewellery-only training set. Meta's official implementation supports promptable image segmentation and provides the model checkpoints used by this project: [SAM 2](#).
- **GrabCut** refines uncertain borders in a narrow band around the initial silhouette. This limits destructive changes to the interior of an already plausible mask.

- **OpenCV consistency rules** keep shadows disjoint from jewellery, keep visible stones inside amodal stones, keep metal inside the jewellery foreground, and measure component counts and boundary support.

Measured results:

Test	Result	Correct interpretation
Independent real-photo sample	SAM score 0.925781	Model confidence only; no human mask exists for accuracy scoring
Real-photo GrabCut refinement	Area ratio 1.000649 relative to initial mask	Conservative boundary refinement, not proof of pixel accuracy
Real-photo support proposals	Four proposals, scores 0.593750 to 0.820312	Possible supports; they are not automatically confirmed as prongs
Ring01 Phase 2.1	One closed amodal stone in every view; front/top contain four prong regions	Structural consistency passed
Ring01 Phase 2.2	Jewellery boundary-edge support 0.92694 to 0.98482	Strong edge agreement with machine evidence, not ground truth
Ring01 Phase 2.2	Front/top stone normalized dimension deltas 0.0268 and 0.03774	Cross-view consistency passed; scale remains unknown

The angled and back views demonstrate the limitation of per-image connected-component counting: occlusion and reflections produced seven and one visible prong fragments respectively. Identity must therefore come from cross-view association and human review, not one-view blob count.

3.3 Dataset segmentation experiment

The only current dataset contains 24 objects with five views each. The split is object-level—18 train, 3 validation and 3 test objects—so different views of one ring never leak across splits.

A 66,229-parameter U-Net was trained on the 18 training objects. It ran on CUDA for 16 epochs at 256-pixel resolution with a batch size of four.

Metric	Result
Training loss	0.647788 to 0.140534
Validation pseudo-mask IoU	0.978183
Held-out test pseudo-mask IoU	0.977586
Held-out test Dice	0.988646
Held-out test boundary F1 at 2 px	0.999753

These are reproduction scores against OpenCV-derived pseudo-silhouettes. They demonstrate that a small model can learn the current normalization target. They do **not** demonstrate 97.8% jewellery understanding, component accuracy or 3D accuracy.

3.4 Phase 2.3: universal detail evidence and review gate

Phase 2.3 was introduced because fixed labels such as “stone” and “prong” are not sufficient for idols, faces, filigree, reliefs, chains or unknown ornaments.

The stage:

- preserves the original image and immutable machine evidence;
- proposes local instances without forcing a jewellery label;
- assigns stable observation IDs and cross-view track hypotheses;
- supports `accept`, `reject` and `relabel` decisions;
- materializes reviewed component masks separately from machine evidence;
- blocks geometry handoff if reviews are incomplete or cross-view identities conflict.

This prevents a bright reflection from silently becoming a gemstone or a shadow from becoming a cavity in CAD.

3.5 Phase 3: multi-view geometry

The dataset-wide coarse reconstruction uses aligned orthographic silhouettes, voxel carving, marching cubes and Trimesh export.

Result	Value
Objects reconstructed	24
Visual-hull resolution	128 voxels
Train reprojection IoU	0.948832
Validation reprojection IoU	0.946122
Test reprojection IoU	0.961753
Watertight meshes	24 / 24
Readable STL and 3MF exports	24 / 24

These high values show that each hull reprojects to its own input silhouettes. A visual hull cannot recover hidden concavities, separate stones from metal or prove scale, so these models remain explicitly named `non_metric`.

3.6 Ring01 exact-solid refinement

Ring01 uses CadQuery/OpenCascade to construct valid parametric solids and a render-and-compare loop to fit them to all five views. Trimesh verifies derived meshes and export readability.

Checkpoint	Mean silhouette IoU	Detail IoU	Boundary F1	Decision
Phase 3 coarse	0.744920	—	—	Useful initialization
Phase 3.2	0.590510	—	0.466620	Retained as a documented intermediate failure
Phase 3.3	0.818114	0.784277	0.795217	Accepted exact-solid checkpoint
Phase 3.3.1	0.819726	0.787388	0.795969	Accepted smooth, individually named prongs
Phase 3.3.2	Same as Phase 3.3.1	Same	Same	Accepted inspection-camera correction; geometry unchanged

Phase 3.3.1 also passes these topology checks:

- one connected, closed and valid metal B-rep;
- one separate, closed and valid gemstone B-rep;
- four curved cubic prongs with stable IDs;
- open two-rail gallery;
- one shank and four shoulders;
- faceted separate stone;
- watertight derived print mesh.

The per-view silhouette IoUs are front 0.854535, side 0.813976, top 0.871267, angled 0.770454 and back 0.788398. The current weak point is fine semantic agreement: component mean IoU is only about 0.37–0.45, and human masks are unavailable. Phase 3 therefore has a valid checkpoint but has not reached the 0.90 research target or manufacturing accuracy.

4. Depth, texture, edge and fine-detail reconstruction

The reconstruction cannot treat every visible line or colour change as physical shape. Polished metal, gemstones, enamel, shadows and engraving create different image signals. This extension separates base geometry, surface detail and appearance so the optimisation cannot turn a highlight into a groove or permanently bake a reflection into the metal colour.

4.1 Required output layers

The output should contain three independently validated layers:

1. **Base geometry:** watertight metal bodies, settings, galleries, links, stones, seats and cavities.
2. **Microgeometry:** engraving, embossed relief, deity faces, floral motifs, milgrain, ridges and displacement that materially changes the surface.
3. **Appearance:** base colour, material identity, metalness, roughness, normal maps and view-dependent reflective behaviour.

Every visible feature must retain source-view provenance and confidence. Detail supported by several views can become geometry. Detail visible in one reflective view remains a hypothesis until reviewed or supported by a known jewellery pattern.

4.2 Camera, depth and normal ensemble

There is no single trusted depth model. The planned benchmark uses independent branches and keeps their disagreements:

Branch	Proposed algorithms	Project role
Classical multi-view geometry	COLMAP and OpenMVS	Recover inspectable camera poses, sparse structure, dense depth and a classical baseline
Feed-forward multi-view geometry	Depth Anything 3 and VGGT	Predict spatially consistent depth, cameras, point maps and tracks from few views
Dense correspondence	MASt3R / DUST3R	Recover cross-view matches when reflective or low-texture regions defeat ordinary features

Branch	Proposed algorithms	Project role
Monocular depth	Depth Pro and Depth Anything V2 Small	Supply sharp boundary and near/far priors without being treated as ground truth
Surface normals	DSINE	Preserve curved metal, facets and shallow relief that depth alone may smooth away

Successful camera solutions will be aligned into one coordinate frame and scored by source-view reprojection. Depth and normal predictions will be fused only where they agree with segmentation, camera geometry and another view. Reflection/shadow probability, occlusion boundaries and model disagreement reduce confidence.

The fused representation should store:

- camera intrinsics, extrinsics and uncertainty per view;
- depth and normal maps with confidence;
- object/component masks and edge classes;
- a signed-distance, occupancy or TSDF volume;
- observed, inferred and unobserved surface labels;
- source-view support for every reconstructed region.

4.3 Edge classification

Before an edge changes geometry, it must be classified:

Edge type	Geometry treatment
Outer silhouette	Strong camera-specific geometric constraint
Component occlusion	Depth-order and component-placement constraint
Crease, ridge or relief boundary	Candidate curve, normal or displacement constraint
Material/enamel boundary	Texture or material mask unless depth evidence agrees
Specular highlight	Lighting/BRDF evidence; never geometry by itself
Cast/contact shadow	Weak spatial cue; excluded from object silhouette
Gemstone facet line	Parametric cut evidence combined with multi-view reflection-aware reasoning

OpenCV Canny/contours remain the deterministic baseline. TEED can propose learned fine edges, while HQ-SAM/HQ-SAM 2 can refine intricate boundaries. The final edge class and component identity require cross-view support or review.

4.4 Texture and reflective-material reconstruction

The texture path should generate separate physically based maps rather than one flattened photograph:

- base colour or albedo;
- metalness/material class;
- roughness;
- normal map;

- displacement/height where geometry is supported;
- optional ambient-occlusion preview;
- source-view visibility and confidence atlas.

Source images must be exposure- and white-balance-normalised. Texture baking should weight each camera by focus, view angle, occlusion, resolution, highlight probability and agreement with neighbouring views. Incompatible observations are not averaged blindly.

For reflective jewellery, NeRO and nvdiffrrec are research candidates for separating geometry, BRDF, material and lighting. PyTorch3D or nvdiffrast can drive differentiable render-and-compare fitting. These systems supply evidence and appearance parameters; they do not replace exact jewellery CAD.

Metal and gemstones need explicit rules:

- polished-metal highlights belong to lighting/BRDF, not permanent albedo;
- enamel, paint and patina belong to material masks and base colour;
- engraving that affects silhouette, shadows or manufacture becomes geometry/displacement;
- gemstones use a named parametric cut and material, not baked sparkle;
- a facet or motif visible in only one view remains low confidence.

4.5 Five-view feasibility and capture upgrade

Five views can constrain principal silhouettes, component layout, broad visible depth, negative spaces and medium detail. They cannot prove hidden surfaces, absolute millimetres, internal wall thickness, gemstone pavilions or physically correct reflective materials.

Five-view input remains supported, but the preferred capture for high-detail validation is:

- the same physical object in every frame;
- fixed focal length, focus, exposure and white balance;
- one measured dimension or scale marker;
- neutral background and substantial overlap;
- front, back, sides, top, underside and oblique coverage;
- 12–36 turntable frames when available;
- macro images for faces, idols, engraving and stone settings;
- a cross-polarised diffuse pass to reduce glare;
- a reflective-light pass for material estimation.

The pipeline should request more views automatically when the five-view confidence map shows unobserved regions or cross-view disagreement.

4.6 Depth and texture validation gates

Layer	Required validation
Camera	Reprojection error, focal plausibility, pose consistency and correct view ordering
Depth	Cross-view agreement and AbsRel/RMSE against scan or CAD where available
Normals	Mean/median angular error and edge-aware normal error
Geometry	Per-view silhouette/boundary scores plus Chamfer, Hausdorff or signed scan-to-CAD deviation

Layer	Required validation
Components	Correct count, stable IDs, pose/dimension errors and no merged stones/prongs
Texture	Photometric metrics, seam error, view consistency and baked-highlight test
Material	Neutral relighting test proving reflections are not fixed into albedo
Uncertainty	Observed, inferred and unknown surfaces reported separately

The benchmark must include five-view and dense-view versions of the same measured object. This isolates the loss caused by sparse evidence from the loss caused by the selected model.

5. Problems encountered and what they taught us

5.1 Reflections were mistaken for geometry

Polished metal produces elongated highlights, while gemstones create reflection and refraction patterns that change with viewpoint. Machine masks classified some highlights as physical openings. The accepted correction filled only reviewed elongated highlight regions and preserved actual gallery openings.

Lesson: image intensity cannot directly define solid/empty geometry. Reflection evidence must be stored separately, and ambiguous regions require review or physically based multi-view reasoning.

5.2 One camera model did not fit independently framed references

A shared camera-distance experiment reduced the proxy objective from the accepted 0.561208 to 0.554328, and front IoU fell to 0.532439.

Lesson: product renders may be independently cropped or rendered. Each view needs its own camera hypothesis unless a calibrated capture proves shared intrinsics.

5.3 A better proxy score produced worse exact CAD

Unconstrained fitting to automatically extracted prong instances increased its local objective but reduced exact mean silhouette IoU to 0.785103, detail IoU to 0.733825 and boundary F1 to 0.759235. It also exported five detached metal solids.

Lesson: machine instance masks are diagnostic proposals. Topology constraints and exact exported-CAD validation must outrank an optimizer's internal score.

5.4 Segmented curved tubes failed at Boolean seams

Piecewise cone segments approximated curved claws visually but created an invalid Boolean seam.

Lesson: smooth multi-section lofts are preferable for claws and organic supports because they preserve both curvature and B-rep validity.

5.5 A preview made an existing prong look missing

The first isometric camera aligned with a 45-degree prong axis, projecting one claw through the gemstone centre. The STEP already contained four named claws.

Lesson: geometry validation and presentation-camera validation are separate. Inspection views must be chosen so important parts do not occlude one another.

5.6 High 2D scores can hide unresolved 3D structure

Visual-hull reprojection exceeds 0.94 on the dataset, yet no camera calibration, physical scale, individual component truth or hidden-surface truth exists.

Lesson: every report must identify what its metric measures. Silhouette agreement, semantic accuracy, 3D surface error, topology validity and manufacturing readiness are different claims.

5.7 The dataset is small and incompletely labelled

The current 120 images represent only 24 independent objects. They have no paired CAD, camera parameters, dimensions, individual stones, seats or material labels.

Lesson: use this dataset for preprocessing, self-supervised multi-view experiments and qualitative stress tests. It cannot train or evaluate exact image-to-CAD reconstruction by itself.

6. Next implementation programme

Milestone A: establish defensible ground truth

This is the immediate priority because additional optimization against unreviewed masks will only overfit machine errors.

1. Human-review all Ring01 Phase 2.3 instances and cross-view tracks.
2. Draw human foreground, metal, stone, setting, prong and negative-space masks for all five views.
3. Provide at least one known dimension—preferably stone diameter and inner-ring diameter.
4. Add camera metadata where available; otherwise record camera uncertainty explicitly.
5. Re-run Phase 3.3 against human evidence and report both old-machine-mask and new-ground-truth scores.

Exit gate: no unresolved review conflicts; five human masks; one metric dimension; reproducible exact-solid report.

Milestone B: individual component and dependency engine

Create a category-independent scene graph rather than a ring-only schema.

Each component should contain:

- stable `component_id` and cross-view `track_id`;
- semantic class and confidence;
- visible and amodal masks per view;
- parent/support relationships;
- pose, dimensions and uncertainty;
- geometry generator or free-form surface reference;
- manufacturing role: metal, stone, cavity, seat, support, decoration or unknown;

- dependencies describing what must update when a parameter changes.

For every stone instance, generate a separate gemstone solid and a matched seat tool. The cavity should be a Boolean subtraction of the transformed seat tool from the metal body, with configurable girdle, pavilion, clearance and wall-thickness allowances.

Exit gate: a synthetic ten-stone test can remove, replace and resize any one stone; only its dependent seat/support geometry changes; all component IDs remain stable.

Milestone C: camera, depth and multi-view correspondence benchmark

Run several geometry estimators on the same reviewed inputs rather than committing to one model:

1. COLMAP baseline for feature-based camera pose and sparse/dense reconstruction.
2. VGGT for feed-forward camera, depth, point-map and track hypotheses.
3. DUS3R/MAS3R as research-only comparison for weakly textured or uncalibrated pairs.
4. Depth Anything V2 Small for a laptop-sized relative-depth prior.
5. DSINE for surface-normal evidence around reliefs and curved metal.

All predictions must be transformed into the project evidence schema with confidence and source provenance. They do not directly overwrite CAD.

Exit gate: calibrated or reviewed camera hypotheses, cross-view track consistency, and lower held-out reprojection/boundary error than the current baseline without breaking topology.

Milestone D: free-form and sculptural detail

Parametric primitives suit shanks, stones and standard settings, but faces, idols, floral relief and hand-carved patterns require a second representation.

The proposed hybrid is:

- exact B-rep for functional and editable jewellery structure;
- local subdivision/NURBS or dense surface patches for organic relief;
- a named attachment frame that binds each patch to the B-rep;
- depth/normal/photometric evidence for the visible relief;
- symmetry only where supported by evidence;
- uncertainty masks on unobserved sides.

Blender can be used to prototype sculpting, remeshing and Boolean cutters. Accepted geometry should return to a validated exchange representation; a Blender Boolean success alone is not sufficient manufacturing validation.

Exit gate: relief details survive render-and-compare, are attached without floating geometry, and do not invalidate the base solid or minimum wall thickness.

Milestone E: manufacturing validation

Add configurable rules for:

- metric scale and tolerances;
- watertight/manifold solids and self-intersections;
- minimum wall and prong thickness;

- stone-seat clearance and girdle contact;
- cavity breakthrough and trapped volumes;
- disconnected or floating metal;
- stone/metal and stone/stone collisions;
- casting, polishing and print allowances;
- mass/volume estimates by material.

The initial rules should be treated as project configuration and reviewed by a jewellery CAD/manufacturing specialist. They must not be invented from images.

Exit gate: a reviewed checklist passes on the authoritative STEP, and physical measurements or scans validate critical dimensions.

7. Algorithms and open-source projects

The project should use an ensemble: deterministic vision for measurable evidence, learned models for ambiguous perception, multi-view geometry for consistency, and exact CAD kernels for editable solids. No single model is expected to solve all four tasks.

7.1 In use now

Project or algorithm	Current use	Why it fits
OpenCV	Normalization, GrabCut, Canny, morphology, contours, components, edges and metric computation	Fast, deterministic, inspectable and CPU-friendly
SAM 2	Prompted whole-object and local proposal masks	Strong general-purpose segmenter with a Tiny checkpoint that fits the available GPU
Small U-Net	Dataset silhouette learning	Only 66,229 parameters; establishes a reproducible task-specific baseline
Visual hull / voxel carving	Coarse multi-view volume	Preserves silhouettes and gives a conservative initialization without pretending to infer hidden concavities
Marching cubes	Converts carved occupancy into a mesh	Standard deterministic surface extraction from a voxel field
CadQuery and OpenCascade	Parametric solids, lofts, Booleans and STEP export	Produces editable exact B-rep geometry instead of only triangle meshes
Trimesh	Mesh loading, validation, voxelization, smoothing and export checks	Lightweight Python geometry toolkit suitable for regression tests
Render-and-compare optimization	Fits CAD and camera parameters to masks	Keeps the final score tied to independently rendered exported geometry

PyTorch3D is present in the research environment and remains useful for differentiable rendering, camera transforms and mesh losses, but the current reported Phase 3.3 results should be attributed to the repository's exact-solid render-and-compare path, not to PyTorch3D.

7.2 Highest-priority additions

Project	Proposed role	Benefits and limits	Integration priority
Grounded SAM 2 + Grounding DINO	Text-prompted detection followed by instance segmentation	Helps propose many small stones and repeated parts; tiled inference can help dense designs. Proposals still require stable IDs and review	High
HQ-SAM / HQ-SAM 2	Boundary-refinement benchmark	Designed for intricate mask boundaries and offers a light variant; must be benchmarked specifically on reflective prongs and filigree	High
Depth Anything V2 Small	Relative-depth prior	Small checkpoint is Apache-2.0 and realistic for the laptop; relative depth still needs cross-view alignment and metric calibration	High
COLMAP	Calibrated-capture SfM/MVS baseline	Mature and measurable; polished, textureless jewellery can defeat feature matching, so controlled diffuse/cross-polarized capture is important	High
VGGT	Camera, depth, point maps and tracks	Useful feed-forward hypothesis generator. Verify the exact checkpoint license before any commercial use	High, benchmark first
PyTorch3D	Differentiable multi-view fitting	Enables silhouette, edge, depth, normal and camera losses in one optimization loop	High
Blender	Synthetic jewellery data, organic detail, Boolean experiments and physically based rendering	Excellent research workbench; accepted solids must still pass CAD/topology validation	High
Manifold	Robust mesh Booleans and manifold repair	Useful for derived print meshes and stress-testing complex cutters; it does not replace exact STEP construction	Medium-high

7.3 Multi-view and reflective-object research candidates

Project	Where it may help	Practical constraint
DUST3R	Pairwise geometry and cameras when classical matching fails	CC BY-NC-SA; research evaluation only unless licensing changes
MASt3R	Dense matching, local features, point maps and global alignment	Non-commercial license and checkpoint-data restrictions require careful review
OpenMVS	Dense point cloud, mesh reconstruction and refinement after camera recovery	AGPL license affects integration architecture; reflective surfaces remain difficult
Meshroom / AliceVision	Reproducible external photogrammetry baseline	Best with many overlapping, sharp images; five studio views may be insufficient
NeRO	Geometry and BRDF estimation for reflective objects	Strong research match for polished metal, but substantially heavier than the current pipeline
NeRSP	Reflective geometry from sparse polarized images	Requires a changed capture process with polarization; valuable controlled-capture experiment
NeRRF	Reflection/refraction-aware research for transparent or specular objects	Experimental neural representation, not directly editable CAD
SpecGloss-GS	Geometry/material separation for glossy objects	Very recent research; benchmark independently before integration

Project	Where it may help	Practical constraint
SuGaR	Surface-aligned Gaussian splatting and mesh extraction	Useful for dense appearance/detail experiments; mesh output still requires semantic and CAD conversion
Neuralangelo	High-fidelity surface reconstruction from many posed images	Official defaults require at least 24 GB VRAM; unsuitable for the 4 GB laptop except heavily reduced or remote runs
Nerfstudio	NeRF/Gaussian-splatting baselines and mesh/point export	Good visual reconstruction laboratory, not proof of exact hidden surfaces or CAD topology

Neural or Gaussian representations should be used as **evidence and geometric priors**, especially for visible free-form detail. Their mesh output must pass through component recovery, exact-solid reconstruction and the same validation gates as every other candidate.

7.4 Coarse generative 3D candidates

Models such as [TRELLIS](#), [Hunyuan3D-2](#) and [InstantMesh](#) may provide a coarse semantic starting shape or novel-view prior. They should not be used as authoritative jewellery CAD because generated meshes can invent hidden details, merge components and violate seat, prong and wall-thickness constraints.

Their correct role is optional initialization:

1. generate a coarse hypothesis;
2. segment it into reviewed components;
3. fit parametric/free-form geometry to real image evidence;
4. discard any unsupported generated detail;
5. rebuild and validate the final CAD independently.

7.5 Jewellery-specific references

There is no mature open-source model found that converts arbitrary jewellery photographs directly into manufacturing-accurate, editable CAD. The closest public work is useful as reference material rather than as a complete replacement:

Project or paper	Useful idea	Limitation for this project
Castable	Parametric ring families, OpenCascade/build123d construction, STEP/STL and casting-oriented validation	Very new and restricted to a few ring styles; audit code and tests before borrowing patterns
Segment Your Ring	Jewellery-specific SAM/LoRA experiment	Dataset contains only two ring types and the author records failures around reflections and jewellery details
JewelSense	Ring detection, SAM segmentation, tracking and Open3D placement	Primarily AR fitting; it does not reconstruct manufacturing CAD from the observed design
ByzantineCAD paper	Feature-based, parametric construction of complex pierced jewellery	Valuable architecture concept, but not a modern open implementation of image-to-CAD

The lack of a complete open solution supports the project's hybrid approach: combine general vision and 3D research with jewellery-specific constraints and editable CAD generation.

7.6 Useful datasets and benchmarks

Dataset	Possible use	Limitation
Current 24-object jewellery set	Normalization, silhouettes, multi-view self-consistency and stress testing	No CAD, scale, cameras or component truth
Amazon Berkeley Objects	Product-image/3D correspondence and material experiments	General products, not jewellery-specific engineering truth
Fusion 360 Gallery	CAD program and construction-sequence learning	Mechanical CAD distribution, not jewellery images
Thingi10K	Mesh robustness, manifold and printability tests	Uncurated triangle meshes; limited semantic consistency
T-ROBI / ROBI	Reflective and textureless-object perception benchmark	Industrial objects rather than jewellery
PASMVS	Synthetic specular multi-view data with depth, masks, cameras and geometry	Domain adaptation to jewellery is still required
3DReflecNet	Large-scale reflective/transparent/low-texture reconstruction research	New benchmark; verify dataset availability, terms and reproducibility before adoption

The most valuable new dataset will be project-owned synthetic-plus-real jewellery data. Blender can render exact CAD with randomized metal, gemstone optics, HDR lighting, camera poses and backgrounds while exporting perfect masks, depth, normals, component IDs and metric geometry. Real captured pieces and scans are then required to measure the synthetic-to-real gap.

8. Efficient experiment order for the available laptop

The current machine has a 4 GB GPU. It can support the existing OpenCV, SAM 2.1 Tiny, small U-Net, CadQuery and low-resolution visual-hull workflow. The next experiments should be ordered by evidence gained per compute cost:

1. Complete human annotation and metric calibration—high value, negligible GPU cost.
2. Benchmark HQ-SAM and Grounded SAM 2 on cropped head/detail regions.
3. Add tiled small-instance detection and stable cross-view IDs.
4. Test Depth Anything V2 Small and COLMAP on controlled captures.
5. Add PyTorch3D losses to the existing exact-solid optimizer.
6. Build the stone-seat-cavity dependency engine with synthetic CAD tests.
7. Run VGGT, DUS3R/MAS3R, NeRO, Neuralangelo or Gaussian-splatting experiments on a larger remote GPU only after a small evaluation set and success criteria are fixed.

This avoids spending days on a heavy model before the ground truth needed to judge it exists.

9. Validation and acceptance gates

9.1 Visual and structural gate

The current Phase 3 research target is:

Measurement	Minimum	Strong target
Mean silhouette IoU	0.80	0.90+
Every principal view	0.70	0.85+
Angled-view IoU	0.70	0.82+
Mean detail-region IoU	0.75	0.85+
External boundary F1	0.75	0.88+
Stable component identity	100%	100%
Required topology and solid checks	100%	100%

These scores become defensible only after target masks are human-reviewed.

9.2 Component-editability gate

- every stone and support has a stable ID;
- no cross-view identity conflict remains;
- removing a component removes only its own geometry;
- changing a stone updates its dependent seat/cavity and required supports;
- metal and stone remain separate valid solids;
- no floating, duplicate or intersecting components remain.

9.3 Manufacturing gate

- physical scale is calibrated from a known measurement or scan;
- critical dimensions are compared against external measurement;
- hidden geometry is confirmed, scanned or explicitly marked inferred;
- wall, clearance, collision, manifold and casting rules pass;
- authoritative STEP opens successfully in an independent CAD application;
- derived STL/3MF is watertight and dimensionally consistent with STEP;
- a domain specialist approves manufacturing assumptions.

10. How reports will drive development

Every run should create a versioned folder and never overwrite an accepted checkpoint. At minimum it should contain:

```
run_id/
├── input_manifest.json
├── environment.json
├── phase1/
│   ├── evidence.json
│   └── previews/
├── phase2/
│   ├── machine_proposals.json
│   └── masks/
├── phase2_3/
│   ├── evidence_graph.json
│   ├── review_decisions.json
│   └── reviewed_components/
├── phase3/
│   ├── fit.json
│   ├── validation.json
│   ├── comparisons/
│   └── model.step
├── manufacturing_report.json
└── decision.md
```

Each validation report should record:

- input hashes, source views and code revision;
- model/checkpoint names, versions and licenses;
- device, runtime and peak memory;
- parameters and random seed;
- machine evidence separately from human-reviewed truth;
- per-view and per-component metrics, not only averages;
- topology, solid, export and manufacturing checks;
- calibrated, inferred and unknown quantities;
- comparison-image paths;
- the exact failed gates;
- final decision: `accept`, `reject`, `backtrack` or `research_only`.

An experiment moves forward only when it improves the intended metric without regressing required topology or another principal view. Rejected approaches remain documented with their measurements so they are not accidentally repeated.

11. Immediate next report sequence

1. **Ground-truth report:** reviewed Ring01 masks, component IDs, known dimensions and uncertainty.
2. **Segmentation benchmark report:** SAM 2.1 Tiny versus HQ-SAM versus Grounded SAM 2 on whole jewellery, stones, prongs, negative spaces and fine details.
3. **Camera/depth benchmark report:** COLMAP, VGGT and lightweight depth/normal priors on the same calibrated capture.
4. **Texture/BRDF report:** albedo, material, roughness, normal and displacement reconstruction with reflection-separation tests.
5. **Phase 3.4 report:** exact-solid re-fit against human ground truth with full per-view/component comparisons.

6. **Parametric edit report:** individual stone resize/removal/replacement and automatically regenerated cavity/seat.
7. **Complex-detail report:** one face/idol/relief sample reconstructed with the hybrid B-rep plus free-form surface path.
8. **Manufacturing report:** metric and topology validation, followed by independent CAD inspection and, when available, scan-to-CAD deviation.

12. Definition of success

The project succeeds when a new jewellery design can pass through the same evidence, review, reconstruction and validation system without adding design-specific hard-coded assumptions.

A successful result must be:

- visibly faithful in every supplied view;
- structurally correct and individually editable;
- honest about hidden and uncertain geometry;
- metric only when physical scale is known;
- reproducible from versioned inputs, code and reports;
- valid as exact CAD before it is described as manufacturing-ready.

The roadmap therefore prioritizes better ground truth and component structure before larger generative models. Better models can improve proposals, but near-perfect jewellery reconstruction will come from combining controlled evidence, multiple algorithms, explicit human review, jewellery constraints, exact CAD and independent validation.